Early Prediction of Student Dropout and Academic Success Using Machine Learning

Group No: 14 — Course: CSE427

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***Abstract*—*Student dropout is an important challenge in higher education because identifying students who are at risk at an early stage can support timely intervention. This study investigates machine learning approaches for predicting student academic outcomes using the Predict Students' Dropout and Academic Success dataset. The Enrolled category is excluded so that the task becomes binary classification between Dropout and Graduate. The workflow includes exploratory data analysis, feature selection, categorical encoding, numerical standardization, stratified train-test splitting, model development, and hyperparameter tuning. Logistic Regression, Decision Tree, Random Forest, XGBoost, Support Vector Machine, and Multilayer Perceptron models are considered. Performance is evaluated using accuracy, weighted F1-score, and ROC-AUC. Based on the test-set results, tuned XGBoost achieved the highest accuracy (0.9050) and weighted F1-score (0.9035), while tuned SVM achieved the highest ROC-AUC (0.9476).*

***Index Terms*—*student dropout prediction, machine learning, higher education, classification, support vector machine, multilayer perceptron, hyperparameter tuning, ROC-AUC*

I. INTRODUCTION

Student dropout in higher education can negatively affect students, institutions, and educational planning. Early identification of students who are likely to discontinue their studies can enable universities to provide targeted academic and administrative support. Machine learning provides a data-driven approach for identifying patterns associated with student outcomes.

Previous studies have investigated dropout prediction using demographic, academic, socioeconomic, and institutional variables. Realinho et al. introduced the dataset used in this work through the UCI Machine Learning Repository [1]. Aulck et al. investigated student dropout prediction in higher education [2]. Kim et al. examined university dropout prediction and associated factors using machine learning techniques [3]. Vaarma and Li presented an empirical study of student dropout prediction in Finnish higher education [4]. Yang investigated academic risk prediction using interpretable machine learning [5].

This project focuses on binary early-outcome prediction. The Enrolled class is removed from the original dataset, leaving Dropout and Graduate as the prediction classes.

II. RELATED WORK

Research on student dropout prediction has explored a wide range of machine learning algorithms, feature groups, and explainability techniques. The following studies were reviewed to position this work within the existing literature.

1) Aulck et al. investigated student dropout prediction using demographic and transcript data from more than 32,500 university students [2]. Several classical classifiers were compared, and the study showed that machine learning could predict attrition with reasonable accuracy using academic information from just a single academic term, without waiting for a full multi-year record. A limitation of this study is its reliance on transcript data from one institutional dataset, which may limit generalizability to other universities.

2) Realinho et al. introduced the dataset used in this project, containing 4,424 students and 36 features covering demographic, socioeconomic, enrollment, and academic information collected at a Portuguese higher-education institution, with Dropout, Enrolled, and Graduate as target classes [1]. The dataset has since been widely adopted as a benchmark for dropout-prediction research. A limitation is that the data originate from a single higher-education institution and exhibit class imbalance across the three outcome categories.

3) Kim et al. examined university dropout prediction using academic, demographic, socioeconomic, and macroeconomic variables and compared several machine learning classifiers [3]. Random Forest achieved the strongest performance, and academic information was found to have the highest impact on prediction accuracy among the feature groups considered. A limitation is that predictions made without academic information were considerably weaker, suggesting the approach depends heavily on the availability of in-program performance records.

4) Vaarma and Li studied dropout prediction using transcript, demographic, and Learning Management System (LMS) data from a Finnish university, evaluating three machine learning models on a monthly basis rather than only at the

end of a semester [4]. Accumulated credits, number of failed courses, and Moodle activity count were identified as the most important predictors, indicating that LMS engagement data adds predictive value beyond transcript and demographic information alone. A limitation is that detailed LMS information may not be available at every institution, and the data came from a single Finnish university.

5) Yang proposed an interpretable machine learning approach using LightGBM combined with SHapley Additive exPlanations (SHAP) to identify academic risk factors and explain individual predictions [5]. The explainability layer allowed specific factors driving each student's risk score to be surfaced rather than only an aggregate accuracy figure. A limitation is that some social and behavioral features may not be present in standard university datasets, restricting the range of explanations that can be generated.

6) Bettahi et al. compared nine machine learning algorithms, including Logistic Regression, Random Forest, k-Nearest Neighbors, SVM, Naive Bayes, and XGBoost, within a modular, explainability-oriented pipeline for dropout prediction using 17,883 undergraduate records from a Moroccan institution [6]. Their tuned XGBoost model achieved an AUC-ROC of 0.993, an F1-score of 0.911, and a recall of 0.944, and subgroup analysis across credit-load bins was used to support fairness auditing. A limitation is that the dataset came from a single Moroccan institution with characteristics that may differ from the dataset used here.

7) A related study evaluated six baseline classifiers before employing a Light Gradient-Boosting Machine and a hybrid stacked ensemble combining Random Forest, LightGBM, and SVM with Logistic Regression as the meta-learner for dropout prediction, extending the task to a three-level risk categorization (low, medium, high) to support more targeted interventions [7]. A limitation is that results depend on the characteristics of the institutional dataset used and on the stability of the risk thresholds chosen.

8) Another study combined Random Forest with survival analysis to predict higher-education dropout risk using administrative enrollment data from 1,385 bachelor's students in business and economics programs [8]. The Random Forest model achieved an AUC of 0.961, an accuracy of 88.6%, and an F1-score of 88.5%, and place of residence and prior secondary-school program were identified as the strongest predictors of dropout. A limitation is that the dataset was relatively small and institution-specific, and it relied only on basic enrollment data.

9) Lopez-Munoz et al. conducted a systematic review of machine learning methods and risk factors for undergraduate dropout prediction, synthesizing studies published between 2019 and 2024 across eight academic databases [9]. The review highlights persistent gaps in understanding psychosocial factors and in translating predictive models into effective, actionable interventions for at-risk students.

Taken together, these studies show that dropout prediction has been approached with a wide variety of algorithms and feature sets, but that academic performance, socioeconomic status, and institution-specific characteristics consistently emerge as important factors. This work compares several classification techniques, including gradient boosting and neural-network models, under a common preprocessing and evaluation framework, and reports results at both the aggregate and class level.

III. DATASET AND PROBLEM DEFINITION

The dataset is the “Predict Students’ Dropout and Academic Success” dataset published through the UCI Machine Learning Repository [1]. The project notebook contains 4,424 student records and 37 variables. The original target contains Dropout, Enrolled, and Graduate. Enrolled is filtered out before model development, resulting in:

$$Y \in \{Dropout, Graduate\}. \quad (1)$$

Thus, the final task is binary classification between students who drop out and students who graduate.

IV. DATA PREPROCESSING

The workflow includes missing-value checks, duplicate checks, descriptive statistics, class-distribution analysis, and exploratory examination of numerical and categorical variables. The dataset was found to contain no missing values across any of its 37 columns and no duplicate rows, so no imputation or de-duplication was required.

Numerical features are standardized using StandardScaler, while categorical variables are transformed using one-hot encoding. A ColumnTransformer is used to apply preprocessing consistently, producing a processed training set of shape (2904, 234) and a processed test set of shape (726, 234) after encoding.

Correlation among numerical attributes was also examined to understand relationships between features prior to model development.

The data are divided into training and testing subsets using an 80:20 split with random state 42 and stratification. Preprocessing is incorporated into model pipelines to reduce the risk of train-test data leakage.

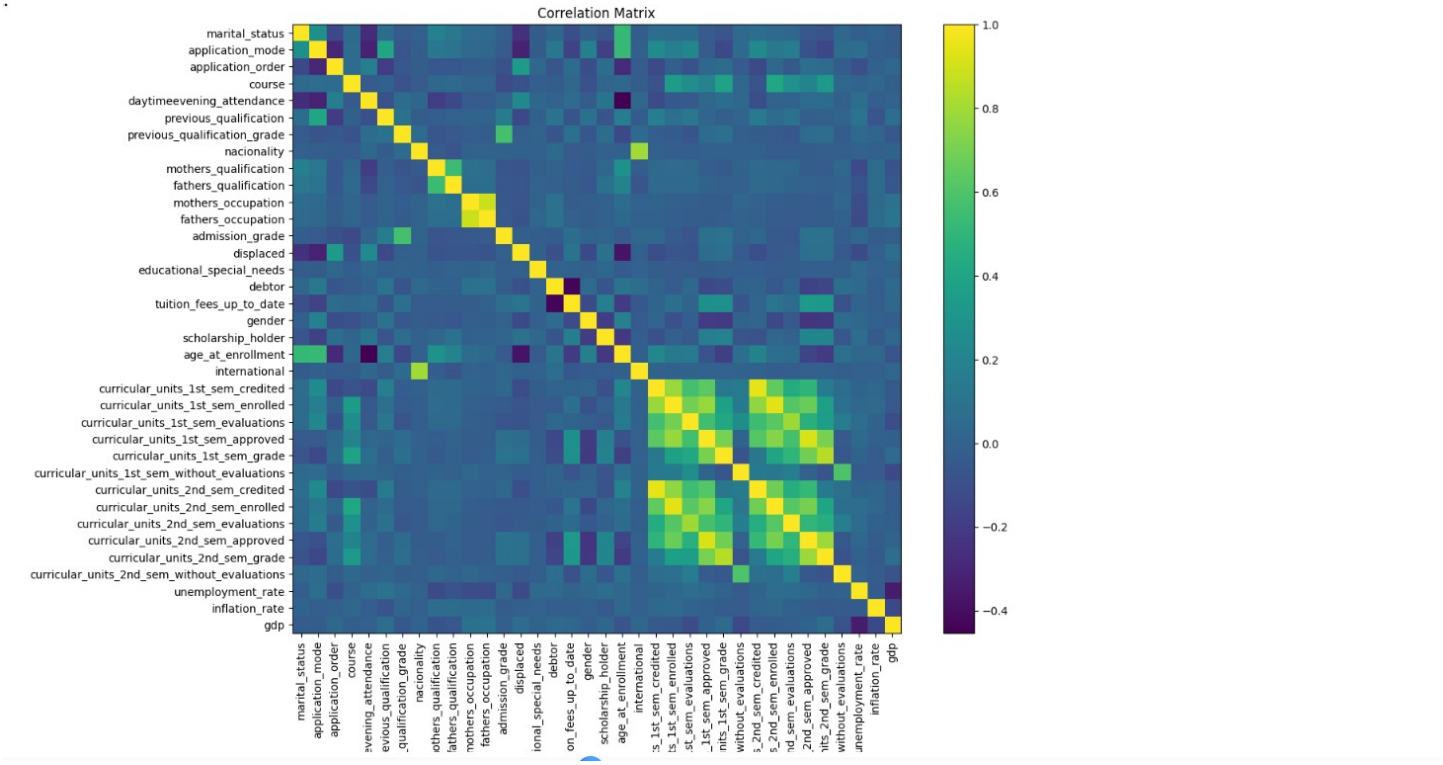


Fig. 1. Correlation matrix of numerical attributes.

V. EXPLORATORY DATA ANALYSIS

The exploratory analysis examines target distribution and relationships between important variables and student outcomes. Numerical variables are compared across outcome groups using box plots, while categorical variables are examined through cross-tabulations. Correlation analysis is also performed for numerical attributes.

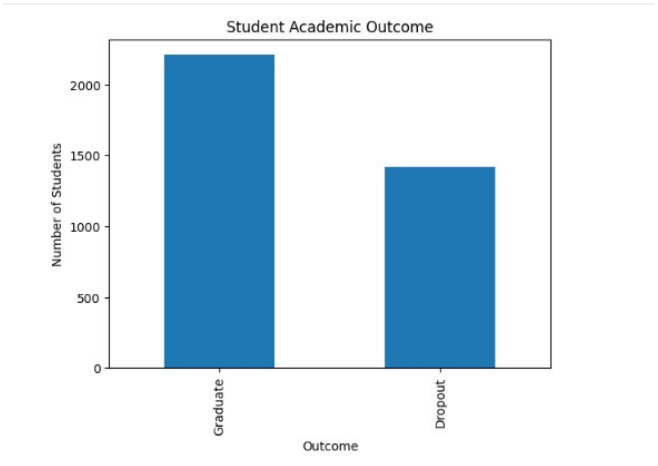


Fig. 2. Distribution of the student academic outcome.

VI. FEATURE SELECTION

The project constructs an early feature set using student background, admission, and selected academic information available under the defined prediction setting. Numerical and categorical features are separated so that appropriate preprocessing can be applied to each group.

VII. MACHINE LEARNING MODELS

Six model families are included in the final comparison.

A. Logistic Regression

Logistic Regression provides a linear baseline and an interpretable reference model.

B. Decision Tree

Decision Tree classification models nonlinear relationships through recursive feature-based splits.

C. Random Forest

Random Forest combines multiple decision trees and aggregates their predictions to improve generalization.

D. XGBoost

XGBoost is a gradient-boosting approach that sequentially constructs an ensemble of decision trees.

E. Support Vector Machine

Support Vector Machine (SVM) identifies a separating decision boundary between the two classes. Kernel, regularization, gamma, and class-weight parameters are considered during tuning.

F. Multilayer Perceptron

Multilayer Perceptron (MLP) is a neural-network classifier using standardized numerical inputs and one-hot encoded categorical inputs. Hidden-layer architecture, activation, learning rate, and regularization are considered during tuning.

VIII. HYPERPARAMETER TUNING

GridSearchCV with five-fold cross-validation is used for hyperparameter optimization. Weighted F1-score is used as the primary tuning metric. For SVM, the regularization parameter C, kernel, gamma, and class-weight strategy are tuned. For MLP, hidden-layer architecture, activation function, learning rate strategy, and L2 regularization are considered. The selected models are subsequently evaluated on the held-out test set.

IX. EVALUATION METRICS

Three primary metrics are used.

A. Accuracy

Accuracy measures the proportion of correctly classified observations:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN). \quad (2)$$

B. F1-Score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \cdot (Precision \times Recall) / (Precision + Recall). \quad (3)$$

Weighted F1-score is used for the overall binary classification comparison.

C. ROC-AUC

ROC-AUC measures the ability to distinguish the two classes across classification thresholds.

X. RESULTS

The test-set results show that tuned XGBoost achieved the highest accuracy (0.9050) and weighted F1-score (0.9035). Tuned SVM achieved the highest ROC-AUC (0.9476), while MLP achieved an ROC-AUC of 0.9260. Decision Tree and Random Forest both achieved an accuracy of 0.8871.

TABLE I
TUNED MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8967	0.8978	0.8967	0.8955	0.8818
Decision Tree	0.8871	0.8897	0.8871	0.8851	0.8682
Random Forest	0.8871	0.8906	0.8871	0.8849	0.8670
XGBoost	0.9050	0.9077	0.9050	0.9035	0.8880
SVM	0.8994	0.8992	0.8994	0.8989	0.9476
MLP	0.8829	0.8825	0.8829	0.8823	0.9260

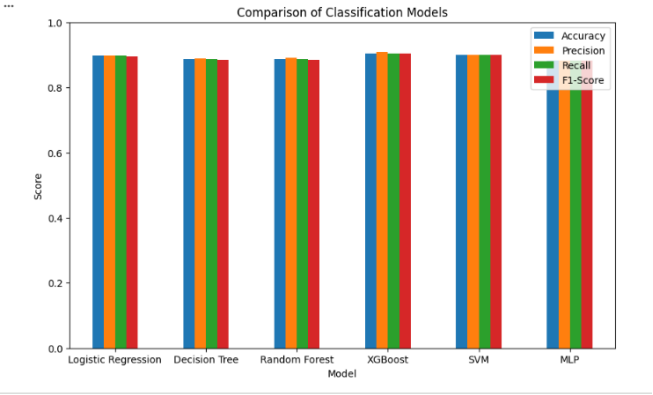


Fig. 3. Comparison of classification models across accuracy, precision, recall, and F1-score.

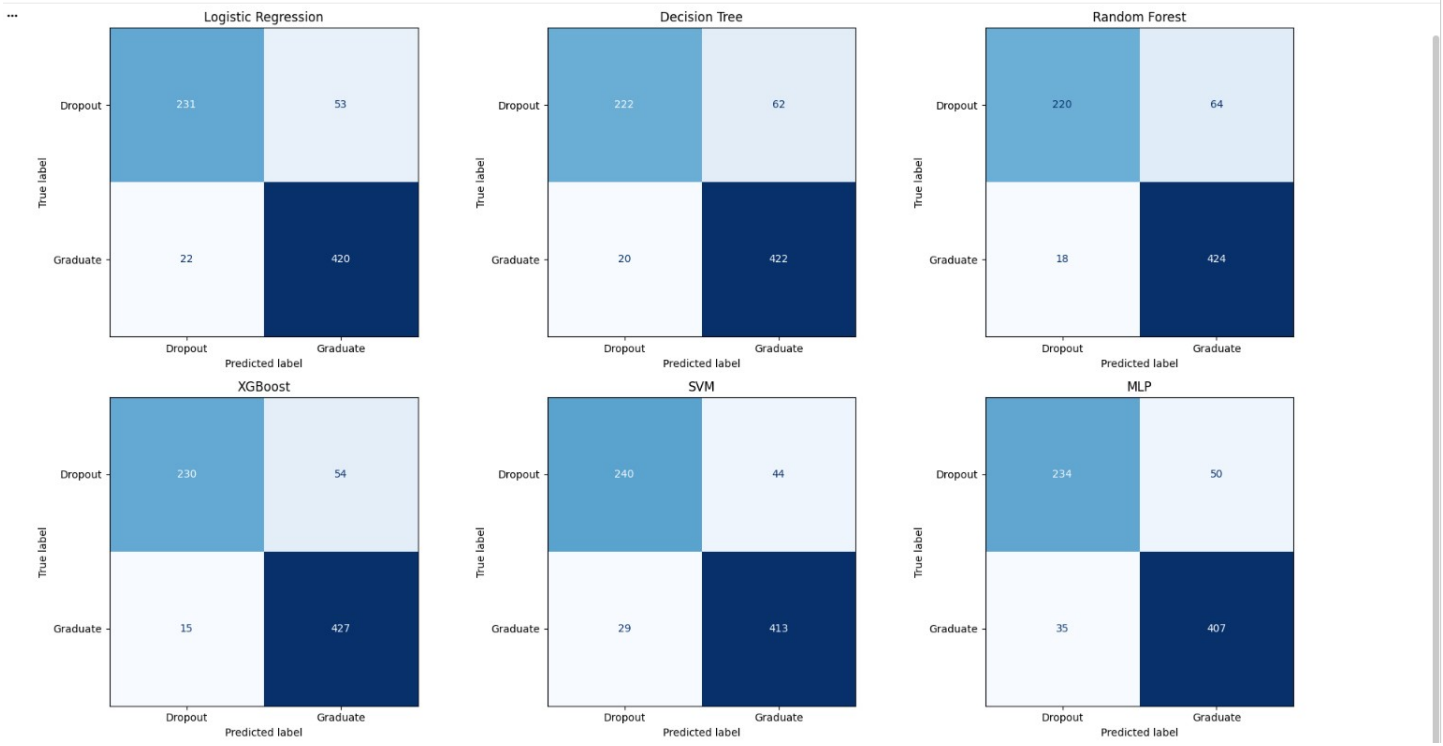


Fig. 4. Confusion matrices for the six tuned models on the held-out test set.

The per-class error patterns for all six tuned models on the held-out test set were also examined. Across all models, Graduate is predicted more reliably than Dropout, which is consistent with the class-level recall values reported below. Tuned SVM produced the fewest false negatives for Dropout (44), followed closely by XGBoost (54) and MLP (50), while Decision Tree and Random Forest produced the most false negatives for Dropout (62 and 64, respectively). This indicates that SVM and XGBoost are more effective at correctly identifying students who actually drop out, whereas the tree-based models without boosting are more prone to misclassifying at-risk students as Graduate.

A. Logistic Regression

The tuned Logistic Regression model achieved 0.8967 accuracy, 0.8978 weighted precision, 0.8967 weighted recall, 0.8955 weighted F1-score, and 0.8818 ROC-AUC. For Dropout, precision, recall, and F1-score were 0.91, 0.81, and 0.86. For Graduate, precision, recall, and F1-score were 0.89, 0.95, and 0.92.

B. Decision Tree

The tuned Decision Tree achieved 0.8871 accuracy, 0.8897 weighted precision, 0.8871 weighted recall, 0.8851 weighted F1-score, and 0.8682 ROC-AUC. For Dropout, precision, recall, and F1-score were 0.92, 0.78, and 0.84. For Graduate, precision, recall, and F1-score were 0.87, 0.95, and 0.91.

C. Random Forest

The tuned Random Forest achieved 0.8871 accuracy, 0.8906 weighted precision, 0.8871 weighted recall, 0.8849 weighted F1-score, and 0.8670 ROC-AUC. For Dropout, precision, recall, and F1-score were 0.92, 0.77, and 0.84. For Graduate, precision, recall, and F1-score were 0.87, 0.96, and 0.91.

D. XGBoost

The tuned XGBoost model achieved 0.9050 accuracy, 0.9077 weighted precision, 0.9050 weighted recall, 0.9035 weighted F1-score, and 0.8880 ROC-AUC. For Dropout, precision, recall, and F1-score were 0.94, 0.81, and 0.87, respectively. For Graduate, precision, recall, and F1-score were 0.89, 0.97, and 0.93, respectively.

E. SVM

The tuned SVM model achieved 0.8994 accuracy, 0.8992 weighted precision, 0.8994 weighted recall, 0.8989 weighted F1-score, and 0.9476 ROC-AUC. For Dropout, precision, recall, and F1-score were 0.89, 0.85, and 0.87, respectively. For Graduate, precision, recall, and F1-score were 0.90, 0.93, and 0.92, respectively.

F. Multilayer Perceptron

The tuned MLP achieved 0.8829 accuracy, 0.8825 weighted precision, 0.8829 weighted recall, 0.8823 weighted F1-score, and 0.9260 ROC-AUC. For Dropout, precision, recall, and F1-score were 0.87, 0.82, and 0.85. For Graduate, precision, recall, and F1-score were 0.89, 0.92, and 0.91.

XI. DISCUSSION

The results indicate that tuned XGBoost provides the highest accuracy and weighted F1-score among the evaluated models, with 0.9050 accuracy and 0.9035 weighted F1-score. Tuned

SVM has the highest ROC-AUC at 0.9476, followed by MLP at 0.9260. Although XGBoost performs best on accuracy and weighted F1-score, SVM demonstrates the strongest overall ranking ability across classification thresholds.

The class-level results show that Graduate recall is higher than Dropout recall for all reported models. This indicates that identifying students who ultimately drop out remains more difficult than identifying students who graduate. The MLP achieved Dropout recall of 0.82, higher than the 0.78 and 0.77 values obtained by the Decision Tree and Random Forest, respectively.

XII. CONCLUSION

This project presents a machine learning framework for predicting student dropout versus graduation. The Enrolled category is excluded so that the final task is binary classification. Logistic Regression, Decision Tree, Random Forest, XGBoost, SVM, and MLP are considered, with hyperparameter tuning performed using five-fold cross-validation.

Based on the test-set results, tuned XGBoost achieved the strongest accuracy and weighted F1-score, with 0.9050 accuracy and 0.9035 weighted F1-score. Tuned SVM achieved the highest ROC-AUC at 0.9476, while MLP achieved 0.9260 ROC-AUC and Dropout recall of 0.82. These results show that model selection can depend on the evaluation metric and the practical objective of the prediction task. Future work can investigate interpretability, class-specific error analysis, probability calibration, and external validation.

ACKNOWLEDGMENT

The authors acknowledge the use of the Predict Students' Dropout and Academic Success dataset made available through the UCI Machine Learning Repository.

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